

2.1:

- Seems fine. Need to see later relevance to know if its "sufficient". (Extra comment in U_F !)

2.1.1: For Hamiltonian dynamics, Q is conserved by the Hamiltonian equation

$$\dot{Q} = [H, Q] = 0$$

Fluctuation: $U_F^\dagger Q U_F = Q \Rightarrow [Q, H_F] = 0$

* $\beta = \frac{1}{k_B T}$, $T \rightarrow \infty$ $\beta \rightarrow 0$

$$p_\beta = \frac{1}{Z}, \quad Z = \text{Tr}(e^{-\beta H})$$

$$Z = \sum_n \langle n | e^{-\beta H} | n \rangle$$

$$= \sum_{n=1}^N e^{-\beta E_n}$$

$$Z \xrightarrow{\beta \rightarrow 0} \sum_{n=1}^N 1 = 2^N \Rightarrow p_\infty = \frac{1}{2^N}$$

$$\bullet \langle O \rangle_\infty = \text{Tr}^{n=1}(p_\infty O) = \frac{1}{2^N} \text{Tr}(O)$$

2.2:

- Seems fine. Again can reassess when we look at the later sections

2.3: Schrodinger Misspelling

- Statement on Schr. picture not quite right.

Idea:

States evolve, $|\psi(t)\rangle \in \mathcal{H}$

Heisenberg: $|\psi\rangle$, $H(t)$

↳ Depending on the School of thought the opposite view is held:

A state in the Schrodinger picture

Satisfies SCL. eg. and the wave
 $\psi = e^{-iHt}$ is a formal solution to this

2.3.1.: $\mathcal{Q}$: Whether or not densities
 need to be defined.

2.3.2:

Pauli-string basis: tensor product of single site
 Pauli operators

• $\sigma^0 = I, \sigma^x, \sigma^y, \sigma^z$

Pauli-string: $P_\alpha := \sigma_1^{\alpha_1} \otimes \sigma_2^{\alpha_2} \otimes \dots \otimes \sigma_N^{\alpha_N}$

$\alpha_i \in \{0, x, y, z\}$

$\langle A, B \rangle = \text{Tr}(A^\dagger B)$

$\text{Tr}(P_\alpha P_\beta) = 2^N \delta_{\alpha\beta}$

$\mathcal{O} = \sum_\alpha c_\alpha P_\alpha / \text{Tr}(P_\alpha \mathcal{O}) = \sum_\beta c_\beta \text{Tr}(P_\beta \mathcal{O})$

$\text{Tr}(P_\alpha \mathcal{O}) = c_\alpha 2^N \Rightarrow c_\alpha = \text{Tr}(P_\alpha \mathcal{O}) / 2^N$

$\Rightarrow$ Local operator: Acts on some sites, say:

$\mathcal{O} = I_1 \otimes I_2 \otimes \dots \otimes I_{i-1} \otimes \mathcal{O}_i \otimes I_{i+1} \otimes \dots$

$\Rightarrow$ So perhaps, if used later on write this
 out

$\text{End}(\mathcal{H}) \cong \mathcal{H} \otimes \mathcal{H}^*$

$\hookrightarrow$ all elements are written as $|p\rangle$

• $\dot{p} = -i[H, p] = \mathcal{L}p$

operators

$\frac{d}{dt} \mathcal{O} = i[H, \mathcal{O}]$

$\frac{d}{dt} |\mathcal{O}(t)\rangle = i \mathcal{L}[\mathcal{O}] = i [H, \mathcal{O}]\rangle$

(Maybe earlier)

Quantum transport:

$\nearrow$

local density q_j , define what "this transport" is. maybe also state roughly how q_j is defined

$$Q = \sum_j q_j$$

local continuity

j_i : current from i to $i+1$

$$\frac{dq_i}{dt} = j_{i-1} - j_i$$

$$\frac{d}{dt} \sum_i q_i = \sum_i (j_{i-1} - j_i) = 0$$

lattice version of

$$\nabla^{\text{div}} Q \equiv \partial_t q(x,t) + \partial_x j(x,t) = 0$$

* Notation of (2.1)

$$C_J(t) \equiv \langle J(t) J(0) \rangle$$

Measures the extent to which the current at time t retains information about its value at t_0 .

* Redundancy in sentence pg 14: "overlap"

2.4.1

$$D_{GK}(t) = \frac{1}{N^2} \int_0^t C_J(t') dt'$$

← another common

$$D = \lim_{t \rightarrow \infty} D_{GK}(t)$$

* Non decaying $D_{GK}(t) \sim t : \sum_b \langle j_b(t) j_b(0) \rangle$

$$D_{GK}(t) \sim t$$

$$C_J(t) = \frac{1}{2^N} \text{Tr}(J(t) J(0))$$

$$C_J(t) = \sum_i \langle j_i(t) j_i(0) \rangle = \frac{1}{N} \sum_i \text{Tr}(j_i(t) j_i(0))$$

$$j = \sum_{\alpha} j_{\alpha}$$

$$C_j = \frac{1}{2^N} \text{Tr} [j(t) j(0)] = \frac{1}{2^N} \sum_{\alpha\beta} \text{Tr} [j_{\alpha}(t) j_{\beta}(0)]$$

Sum over β yields all the same so can

Choose any

$$C_j = \frac{1}{2^N} N \sum_{\alpha} \text{Tr} [j_{\alpha}(t) j_{\alpha}(0)]$$

$$= N C_{jj}$$

$\Rightarrow$ So, just define that densities are rep. by their respective lower case.

$$D_{6a}(t) = \frac{1}{\chi} \int_0^t C_{jj}(t') dt'$$

comma?

$$C(p) = \langle j(p) j(0) \rangle$$

$$D_K(p) = \frac{1}{N\chi} \sum_{p'=0}^P C(p') \Rightarrow D_K(p) = \frac{1}{\chi} \sum_{p'=0}^P C_{jj}(p')$$

$$C(x,t) := \langle q_x(t), q_0(0) \rangle = \frac{1}{2^N} \text{Tr} [q_x(t) q_0(0)]$$

$\leadsto$ Don't know if its required to "define" a fluctuation.

$$C(x,t) \sim \sqrt{t}$$

$$2.4.3: \text{MSD}(t) = \frac{\sum_x x^2 C(x,t)}{\sum_x C(x,t)}$$

$$\text{Def: } C(x,t) \sim \sqrt{t} f(x)$$

$$\text{MSD}(t) \sim \sum_x x^2 f(x) / \sum_x f(x)$$

$$D(t) = \frac{1}{2} \frac{d}{dt} \text{MSD}(t)$$

$$\text{MSD}(t) = 2D/t$$

2.5: Models

2.5.1: Mixed Field Ising Mod

$\leadsto$ 1D chain of spin-1/2 D.O.F

$$H = \sum_j h_j = \sum_j (g_x X_j + g_z Z_j) + \frac{1}{2} (Z_{j-1} Z_j + Z_j Z_{j+1})$$

X_j, Z_j : Pauli operators

[Eq 28] I would check this as it seems very particular to your lecture notes/Book of choice. the () is missing but the sign in $H = \pm J \Sigma \dots$

depends on

$J > 0$ $\uparrow\uparrow$ fer.

$J < 0$ $\uparrow\downarrow$ para.

and let's say a constant $J/2 (Z_{j-1} Z_j + Z_j Z_{j+1})$ which characterizes these interactions.

$g_z = 0$: $\mathbb{Z}_2$ symmetry $Z_i \rightarrow -Z_i$

$\hookrightarrow \prod X_i$

Bit flip operator $P = \prod_{i=1}^N X_i$ OK

$$[P, H] = 0 \quad P^2 = \mathbb{I}$$

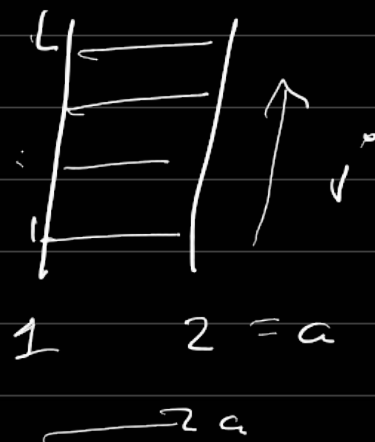
compared to other sectors.

* Lacking detail in Conserved Quantities/Current.

XX Ladder: $a = 1, 2, \dots, L$ rungs

$j = 1, \dots, L$
 X_{aj}, Y_{aj}, Z_{aj}

$$H = \sum_{j=1}^{L-1} \sum_{a=1}^2 (J_{||} \uparrow) + \sum_{j=1}^L J_{\perp} \uparrow$$



~~explain why these coefficients are certain~~

~~if they are not explain why.~~

$J_{\parallel}$: interaction hopping
 $J_{\perp}$: ring hopping coeff

~~Set~~

Set to 1

Also set disorder term to zero $h_i = 0$

$$Q = \frac{1}{2} \sum_{j=1}^L \sum_{a=1,2} Z_{ja} = \sum_{j=1}^L q_j, \quad q_j = \frac{1}{2} (Z_{j1} + Z_{j2})$$

* Refrase ring X_0

$$C(x, t) = \frac{2}{2^N} \text{Tr}[q_{x+x_0}(t) q_{x_0}(0)]$$

$$N = 2L$$

$$x=0: \quad C(0,0) = 1 \Rightarrow \frac{2}{2^N} \text{Tr}[q_{x_0}(0) q_{x_0}(0)] = 1$$

2.5.3 Floquet XXZ Chain

Spin $\frac{1}{2}$ on F in 1D lattice

Time periodic Hamiltonian

$$H(t) = H(t+\tau)$$

$$U(\tau) = \mathcal{T} \exp\left(-i \int_0^\tau H(t) dt\right)$$

U(1) -symm

$$S_a^x = \sigma_a^x / 2$$

$$U = \exp\left[-i \left(J_{xy} (\cdot) + J_{zz} + g (S_1^z - S_2^z)\right)\right]$$

$$\Rightarrow [U, S_1^z + S_2^z] = [U, S_{tot}^z] = 0$$

Notation: Here u is used and before

U_F . Also σ used and X, Y, Z before

$$q_j = S_j^z$$

$$C(x, p) = \frac{1}{2^N} \text{Tr}[q_{x+x_0}(p) q_{x_0}(0)]$$

* Should U have a t value?

- Same notation issues: if $q_j = S_j^z$
need a different notation as here for the
entire system i.e. each two $(U_F)_i$ etc.

2.6: $P_0 = |0\rangle\langle 0|_c$, $P_1 = |1\rangle\langle 1|_c$

$$\begin{aligned} D_c(\rho) &= P_0 \rho P_0 + P_1 \rho P_1 \\ &= |0\rangle\langle 0| \rho |0\rangle\langle 0| \\ &= \langle 0| \rho |0\rangle |0\rangle\langle 0| + \langle 1| \rho |1\rangle |1\rangle\langle 1| \\ &= P_0 + P_1 \end{aligned}$$

$$P_0 = \frac{I_c + Z_c}{2}, \quad P_1 = \frac{I_c - Z_c}{2}$$

$$\frac{1}{4} (I + Z) \rho (I + Z) = \frac{1}{4} (\rho + Z \rho) (I + Z)$$

$$= \frac{1}{4} (\rho + Z \rho + \rho Z + Z \rho Z)$$

$$\begin{aligned} &= (\rho - Z \rho) (I - Z) = (\rho - Z \rho - \rho Z + Z \rho Z) \\ &= \frac{1}{2} (\rho + Z \rho Z) \quad \checkmark \end{aligned}$$

- $\rho = \sum_{ij} p_{ij} |i\rangle\langle j|$

2.7: Not quite just the mean but the
cardinality of the mean

$$|\phi_{ex}\rangle = U(t) |\phi(w)\rangle$$

- Again, whether each state is strictly with same
notation.

2.8: Here $\hat{O}$, not observable

2.6.2: $|n\rangle = |n_b\rangle \otimes |n_w\rangle$

$$|u\rangle = |1-u_{j_0}\rangle \otimes |u_{ew}\rangle$$

$P_n(t) \sim \langle \text{State}(t) \rangle$ | ~~fn~~ missing last eq pg 29.

2.8.3: fine

2.8.4:

* Chapter 4

4.4: All seems fine here, the only thing
is the last sentence really should be
noted in Chapter 2.

4.4.1:

Although you've written it more compactly this hasn't
been through a choice/requirement of your system
so, it might be better to have this in chap
2.

• Don't have references so can't quite check.

4.4.1.1: Again its okay to relabel with a
but just be sure this is common practice

$$H_i H_j = (Z_i Z_{i+1} + \tilde{f}_i + \hat{f}_{i+1}) (Z_j Z_{j+1} + \tilde{f}_j + \hat{f}_{j+1})$$

$$= (Z_j Z_{j+1}) (Z_i Z_{i+1}) + \dots$$

$$\tilde{f}_{i+1} Z_j Z_{j+1} = (g_x X_i + g_z Z_i) (Z_j Z_{j+1})$$

↳ $X_i Z_j \neq 0$ for overlapping sites

$$H \sim H_{ev} \oplus H_{od}$$

any two elements in H_e or H_o have non overlap

sup.

4.4.2: Above had references to why values were chosen, here not so much.

Also explain why one goes from X_i to σ_i^x and vice versa.

Both that & relations they.

4.4.2.1:

$$(X \otimes X + Y \otimes Y) |00\rangle = X|0\rangle \otimes X|0\rangle + Y|0\rangle \otimes Y|0\rangle$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i^2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= |11\rangle - |11\rangle = 0 \quad \checkmark$$

$$(X \otimes X + Y \otimes Y) |01\rangle = X|0\rangle \otimes X|1\rangle + Y|0\rangle \otimes Y|1\rangle$$

$$= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes (-i) \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= |10\rangle - i^2 |10\rangle$$

$$= 2 |10\rangle \quad \checkmark$$

$$|10\rangle = |0\rangle \otimes |1\rangle + (-i) |0\rangle \otimes i |1\rangle$$

$$= 2 |01\rangle \quad \checkmark$$

$$H_B = \begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix} + i \begin{pmatrix} 0 & -\sigma_y \\ \sigma_y & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} : \boxed{\{|01\rangle, |10\rangle\}} \equiv A$$

$$\hookrightarrow = 2\sigma_x \quad \checkmark$$

$$H_B|_A = 2\sigma_x$$

$$\Rightarrow U_b(t) = e^{-i(2\sigma_x)t} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(2t) & -i\sin(2t) & 0 \\ 0 & -i\sin(2t) & \cos(2t) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Not sure if d is meant to represent

Some scaling of t or a coupling.
and d can be introduced if you define

$$H_{\text{bond}} = d(x \otimes x + \gamma \otimes \gamma)$$

But most likely a period.

4.4.3: Same as earlier, give reference to any
material which has guided the choices used or
explain why those are made (later on, the
couplings are fine).

4.4.4: Slight notational difference

$$z_{src} = 2u_{j_0} - 1 \in \{-1, +1\} \quad \text{pg 26 chp 2}$$

$$z_{src} = 1 - 2u_{src} \in \{-1, +1\} \quad \text{pg 6 chp 4}$$

* Maybe I missed it but for

$w_{src} (p_u - p_{\bar{u}})/2$, I don't see where p_u is
defined/referenced. (Probably from chp 2)

* references jc for MFM but not much in sections.

$$C_{jj}(0) = \frac{1}{2N} \text{Tr}(g_x^2 \dot{\phi}^2) = 2g_x^2$$

